

Hackathon Report

Configurable Workflow Decision Platform

1. Problem Statement

Build a resilient decision system to process workflows, evaluate rules, manage state, and handle failures.

2. Solution Overview

A configuration-driven workflow engine using JSON rules to determine outcomes like approved, rejected, manual_review, retry_pending.

3. Tech Stack

Backend: Node.js, Express

Database: MongoDB

Frontend: React (Vite)

4. Architecture

API Layer, Workflow Engine, Rule Engine, Database, Audit Logs, External Dependency Simulator.

5. Workflow Execution

Receive request → validate → idempotency → process → evaluate rules → store result.

6. Features

Validation, Rule Engine, State Management, Audit Logs, Idempotency, Failure Handling.

7. APIs

POST /submit

GET /request/:id

GET /audit

8. Example

Input: valid data → Output: approved

9. Conclusion

System is configurable, scalable, and handles real-world workflow challenges.